

———— // ZPWR-MIDI-FX // ————

zpwr-midi-fx

v0.1.0

MODULE REFERENCE

wire your own MIDI processor from note-stream blocks

2026-06-17

MenkeTechnologies

———— MENKETECHNOLOGIES ————

Signal Sources

External sources you can cable into any block input or modulation route.

1 MIDI In 3 Random Mod Wheel Pitch Bend Aftertouch Velocity Expression Sustain MPE Bend MPE Pressure MPE Slide

Modulation

7 modules

Generative 3 inputs

Probabilistic melodic random walk synced to a rate.

Param	Range	Default	Unit
Rate	$1/1 \cdot 1/2 \cdot 1/4 \cdot 1/8 \cdot 1/16 \cdot 1/32 \cdot 1/4 \cdot 1/8 \cdot 1/16 \cdot 1/4T \cdot 1/8T \cdot 1/16T$	—	—
Range	1 – 12	3	st
Low	0 – 127	48	—
High	0 – 127	72	—
Probability	0 – 1	0.7	—

MidiLF0 3 inputs

Low-frequency modulation source (sine/tri/saw/square) on the scalar control output.

Param	Range	Default	Unit
Rate	0.01 – 20	1	Hz
Shape	Sine · Triangle · Saw · Square	—	—
Depth	0 – 1	1	—

MidiEnv 3 inputs

ADSR envelope triggered by the note gate, on the scalar control output.

Param	Range	Default	Unit
Attack	1 – 5000	5	ms
Decay	1 – 5000	200	ms
Sustain	0 – 1	0.7	—
Release	1 – 5000	300	ms

MidiRandom 3 inputs

Generates random notes within a pitch range at a synced rate.

Param	Range	Default	Unit
Rate	$1/1 \cdot 1/2 \cdot 1/4 \cdot 1/8 \cdot 1/16 \cdot 1/32 \cdot 1/4 \cdot 1/8 \cdot 1/16 \cdot 1/4T \cdot 1/8T \cdot 1/16T$	—	—
Low Note	0 – 127	48	—
High Note	0 – 127	72	—
Gate	0 – 2	0.5	—
Velocity	1 – 127	100	—

MidiSampleHold 3 inputs

Samples and holds the incoming control value on each note.

MidiSlew 3 inputs

Smooths (glides) the control value over a set time.

Param	Range	Default	Unit
Time	0 – 2000	100	ms

RandOctave 3 inputs

Randomly shifts notes by up to +/- Range octaves with a chance.

Param	Range	Default	Unit
Range	1 – 3	1	—
Chance	0 – 1	1	—

Dynamics

6 modules

Velocity 3 inputs

Scales and offsets note velocity, with a modulation amount.

Param	Range	Default	Unit
Scale	0 – 2	1	—
Offset	-64 – 64	0	—
Mod Amt	-64 – 64	0	—

VelCurve 3 inputs

Reshapes velocity through a power curve.

Param	Range	Default	Unit
Curve	0.2 – 5	1	—

Humanize 3 inputs

Adds random timing and velocity jitter for a human feel.

Param	Range	Default	Unit
Timing	0 – 50	8	ms
Velocity	0 – 1	0.3	—

Accent 3 inputs

Boosts velocity on every Nth note.

Param	Range	Default	Unit
Every	1 – 16	4	—
Amount	0 – 1	0.5	—

VelClip 3 inputs

Clamps velocity between Min and Max.

Param	Range	Default	Unit
Min	1 – 127	1	—
Max	1 – 127	127	—

Ramp 3 inputs

Velocity ramp over a run of notes, ascending or descending.

Param	Range	Default	Unit
Length	2 – 16	8	—
Depth	0 – 1	0.6	—
Direction	Up · Down		—

Control

5 modules

CCMap 3 inputs

Remaps a CC number and scales/offsets its value.

Param	Range	Default	Unit
From	-1 – 127	1	—
To	0 – 127	11	—
Scale	0 – 2	1	—
Offset	-1 – 1	0	—

Note2CC 3 inputs

Emits a CC from each note (pitch or velocity).

Param	Range	Default	Unit
CC	0 – 127	1	—
Source	Pitch · Velocity		—

CC2Note 3 inputs

A watched CC crossing a threshold triggers a note.

Param	Range	Default	Unit
Watch	0 – 127	1	—
Note	0 – 127	60	—
Threshold	0 – 1	0.5	—

BendProc 3 inputs

Scales and/or inverts incoming pitch-bend.

Param	Range	Default	Unit
Scale	0 – 2	1	—
Invert	0 – 1	0	—

ATProc 3 inputs

Scales aftertouch, optionally converting it to a CC.

Param	Range	Default	Unit
Scale	0 – 2	1	—
To CC	-1 – 127	-1	—

Param	Range	Default	Unit
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Harmony

13 modules

Chord 3 inputs

Turns each incoming note into a chord (165 types) with inversion, spread, octave doubling and strum.

Param	Range	Default
Type	Unison · Octave · 5th (Power) · 5th + Oct · Tritone · Major · Minor · Diminished · Augmented · Major (1st inv) · Major (2nd inv) · Mi	
Inversion	0 – 4	0
Spread	0 – 2	0
Octave x2	0 – 2	0
Transpose	-24 – 24	0
Strum	0 – 200	0

Scale 3 inputs

Quantizes incoming notes to the nearest degree of the chosen root and scale.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pentatonic M	

Transpose 3 inputs

Shifts every note by a fixed semitone amount plus a modulatable offset.

Param	Range	Default	Unit
Semitones	-24 – 24	0	st
Mod Amt	-24 – 24	0	st

Harmonize 3 inputs

Adds up to three fixed-interval harmony voices to each note.

Param	Range	Default	Unit
Interval 1	-24 – 24	7	st
Interval 2	-24 – 24	0	st
Interval 3	-24 – 24	0	st

Drone 3 inputs

Doubles every note with a fixed pedal root.

Param	Range	Default	Unit
Root	0 – 127	36	—

MidiCluster 3 inputs

Adds symmetric tone-cluster voices above and below each note.

Param	Range	Default	Unit
Width	0 – 4	1	—
Spread	1 – 4	1	st

Glissando 3 inputs

Fills a stepwise run between consecutive notes.

Param	Range	Default	Unit
Rate	5 – 200	40	ms

ChordMem 3 inputs

Each key plays a stored chord shape (semitone intervals from node config).

ChordProg 3 inputs

Each note triggers the next chord in a stored root progression (node config).

ChordVoicing 3 inputs

Re-voices each note into a jazz shape (open / shell / quartal / spread).

Param	Range	Default	Unit
Voicing	Open · Shell · Quartal · Spread	—	—

Octave 3 inputs

Doubles each note up and/or down by whole octaves.

Param	Range	Default	Unit
Up	0 – 3	1	—
Down	0 – 3	0	—

MidiFold 3 inputs

Folds out-of-range notes back inside a Low..High window.

Param	Range	Default	Unit
Low Note	0 – 127	48	—
High Note	0 – 127	72	—

Invert 3 inputs

Mirrors pitch around a pivot note.

Param	Range	Default	Unit
Pivot	0 – 127	60	—

Probability

1 modules

Chance 3 inputs

Probabilistic gate: lets each note through with the set probability.

Param	Range	Default	Unit
Probability	0 – 1	0.8	—
Mod Amt	-1 – 1	0	—

Routing

8 modules

Split 3 inputs

Keyboard split: notes below the split point transpose by Low, the rest by High.

Param	Range	Default	Unit
Split	0 – 127	60	—
Low	-24 – 24	0	st
High	-24 – 24	0	st

Transformer 3 inputs

Rule: notes within a range are transposed and velocity-scaled.

Param	Range	Default	Unit
Low	0 – 127	0	—
High	0 – 127	127	—
Transpose	-24 – 24	0	st
VelScale	0 – 2	1	—

Merge 3 inputs

Passes both note-stream inputs through, merged into one stream.

NoteFilter 3 inputs

Passes only notes inside a key and velocity window.

Param	Range	Default	Unit
Low Note	0 – 127	0	—
High Note	0 – 127	127	—
Low Vel	1 – 127	1	—
High Vel	1 – 127	127	—

Latch 3 inputs

Holds notes after release until the next note arrives.

Channel 3 inputs

Reassigns notes to a fixed MIDI channel.

Param	Range	Default	Unit
Channel	1 – 16	1	—

FixedNote 3 inputs

Forces every note to a single fixed pitch (drum / trigger).

Param	Range	Default	Unit
Note	0 – 127	36	—

KeySwitch 3 inputs

Splits the keyboard at a note; one zone passes, the other is filtered.

Param	Range	Default	Unit
Split	0 – 127	60	—
Low Zone	High passes · Low passes		—

Sequencing

18 modules

Arp 3 inputs

Arpeggiator: cycles held notes through a pattern at a synced rate with gate, octaves, swing and step count.

Param	Range	Default	Unit
Mode	Up · Down · Up/Down · Down/Up · Converge · Diverge · As Played · Random · Chord		—
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		—
Gate	0 – 2	0.5	—
Octaves	1 – 4	1	—
Swing	0 – 0.75	0	—
Steps	1 – 16	16	—

SeqEuclid 3 inputs

Euclidean rhythm gate: spreads Pulses evenly across Steps (rotatable) at a synced rate.

Param	Range	Default	Unit
Pulses	1 – 16	5	—
Steps	1 – 16	8	—
Rotate	0 – 15	0	—
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		—
Gate	0 – 2	0.5	—

Swing 3 inputs

Pushes off-beat eighth-notes later for a swing/shuffle feel.

Param	Range	Default	Unit
Amount	0 – 0.75	0.3	—

PatternGate 3 inputs

Gates note-ons through a 16-step on/off pattern (bitmask).

Param	Range	Default	Unit
Pattern	0 – 65535	21845	—
Steps	1 – 16	16	—

NoteOffDelay 3 inputs

Extends each note's gate by delaying its note-off.

Param	Range	Default	Unit
Delay	0 – 2000	200	ms

Ricochet 3 inputs

Accelerating bouncing retriggers with decaying velocity.

Param	Range	Default	Unit
Bounces	1 – 16	6	—
Decay	0.3 – 0.98	0.7	—
Start	10 – 500	100	ms

StepSeqMidi 3 inputs

Melodic step sequencer - semitone offsets (node config) clock the held root.

Param	Range	Default	Unit
Rate	$1/1 \cdot 1/2 \cdot 1/4 \cdot 1/8 \cdot 1/16 \cdot 1/32 \cdot 1/4 \cdot 1/8 \cdot 1/16 \cdot 1/4T \cdot 1/8T \cdot 1/16T$	—	—
Gate	0.05 – 1	0.5	—
Velocity	1 – 127	100	—

Echo 3 inputs

MIDI delay: repeats notes at a synced time with velocity feedback decay.

Param	Range	Default	Unit
Time	1 – 2000	250	ms
Feedback	0 – 0.95	0.5	—
Repeats	1 – 8	3	—

Flam 3 inputs

Precedes every note with a quiet grace note, delaying the main hit so the grace leads it - the classic drum flam.

Param	Range	Default	Unit
Time	1 – 200	30	ms
Grace Vel	0 – 1	0.5	—

Roll 3 inputs

Drum roll / buzz: retriggers every held note at Rate for as long as it is held, each hit gated to a fraction of the interval.

Param	Range	Default	Unit
Rate	1 – 50	12	Hz
Gate	0.05 – 0.95	0.5	—

Trill 3 inputs

Keyboard trill: while a note is held, alternates rapidly between it and the note Interval semitones above, at Rate.

Param	Range	Default	Unit
Rate	1 – 50	10	Hz
Interval	1 – 12	2	st
Gate	0.05 – 0.95	0.5	—

Mordent 3 inputs

One-shot ornament: each struck note plays principal -> auxiliary -> principal at the attack, then sustains, the single-flick counterpart to the continuous Trill.

Param	Range	Default	Unit
Time	5 – 150	40	ms
Interval	1 – 12	2	st
Direction	Upper · Lower		—

Turn 3 inputs

Gruppetto turn: a four-step ornament at the attack - upper auxiliary, principal, lower auxiliary, then the principal held. Time sets each step, Interval the neighbour distance.

Param	Range	Default	Unit
Time	5 – 150	35	ms
Interval	1 – 12	2	st

Appoggiatura 3 inputs

Leaning grace note: each struck note first sounds a long auxiliary (Interval away) that takes Time from the principal, which then sustains - the expressive long grace, unlike the quick Flam or Mordent.

Param	Range	Default	Unit
Time	20 – 400	120	ms
Interval	1 – 12	2	st
Direction	Upper · Lower		—

Strum 3 inputs

Spreads a chord's notes over time, up or down, like a guitar strum.

Param	Range	Default	Unit
Spread	0 – 200	20	ms
Direction	Up · Down		—

MidiQuantize 3 inputs

Snaps note starts to a synced grid by a strength amount.

Param	Range	Default	Unit
Rate	$1/1 \cdot 1/2 \cdot 1/4 \cdot 1/8 \cdot 1/16 \cdot 1/32 \cdot 1/4 \cdot 1/8 \cdot 1/16 \cdot 1/4T \cdot 1/8T \cdot 1/16T$	—	—
Strength	0 – 1	1	—

SeqRatchet 3 inputs

Retriggeres each note Count times within a synced step.

Param	Range	Default	Unit
Rate	$1/1 \cdot 1/2 \cdot 1/4 \cdot 1/8 \cdot 1/16 \cdot 1/32 \cdot 1/4 \cdot 1/8 \cdot 1/16 \cdot 1/4T \cdot 1/8T \cdot 1/16T$	—	—
Count	1 – 8	4	—
Gate	0.05 – 1	0.9	—

NoteLength 3 inputs

Forces a fixed note duration (gate length).

Param	Range	Default	Unit
Length	1 – 2000	200	ms

Tuning

1 modules

Microtuner 3 inputs

Stretched-octave microtuning via per-note pitch detune.

Param	Range	Default	Unit
Stretch	-50 – 50	0	cents/oct

Voicing

4 modules

MPEConvert 3 inputs

Assigns each note its own rotating channel for per-note MPE expression.

Param	Range	Default	Unit
Voices	2 – 15	15	—

Sustain 3 inputs

Sustain-pedal simulation: while Hold is engaged, note-offs are captured so the notes keep sounding; when Hold drops, every held note releases at once. Hold is a parameter, so a CC, an LFO or automation can play the pedal.

Param	Range	Default	Unit
Hold	Off · On		—

Mono 3 inputs

Monophonic note priority (last / lowest / highest).

Param	Range	Default	Unit
Priority	Last · Lowest · Highest		—

MidiUnison 3 inputs

Doubles each note into multiple voices, optionally spread across channels.

Param	Range	Default	Unit
Voices	1 – 4	2	—
Spread	Same chan · Spread chans		—